# 1. Clinical Question & Core Hypothesis

## Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

## Working Hypothesis

In prostate tumors containing both Gleason Grade Group 5 (GG5; cribriform/fused architecture) and Gleason Grade Group 1 (GG1; well-formed glands) regions within the same patient, the GG5 component exhibits a coherent mechanistic program distinct from GG1. This program is driven by a cascade of molecular events that collectively promote aggression: PTEN loss with activation of PI3K/AKT and upregulated AR signaling; frequent ERG gene fusion; co-occurring TP53 pathway disruption; BRCA2 alterations or homologous recombination deficiency; higher Ki-67 proliferation; and enhanced invasive capacity (perineural/extraprostatic extension). Compared with GG1 regions, GG5 regions will show (i) PTEN loss with PI3K/AKT/mTOR axis activation and increased AR activity, (ii) ERG fusion co-occurring with PTEN loss, amplifying oncogenic signaling, (iii) TP53 mutation or dysfunction linked to genomic instability and invasion, (iv) BRCA2/HR defects contributing to genomic stress and aggressiveness, (v) elevated Ki-67, and (vi) higher incidence of invasion and earlier biochemical recurrence or metastasis. The edges to test include: GG5 architecture -> PTEN loss; PTEN loss -> PI3K/AKT activation; ERG fusion + PTEN loss -> amplified signaling; TP53 mutation -> genomic instability/invasion; BRCA2/HR deficiency -> repair failure -> aggressiveness; Ki-67 elevation -> recurrence risk; cribriform pattern -> higher biochemical recurrence; AR signaling upregulation -> proliferation/survival. This paired Case B framework should reveal a consistent, architecture-linked molecular program that explains the poorer prognosis of GG5 relative to GG1 and provides a mechanistic basis for prognosis and potential therapeutic targeting.

# 2. Why This Gene — Not Others

# 3. Clinically Testable Predictions

**Prediction 1:** Collect radical prostatectomy specimens with coexisting GG1 and GG5 regions;

**Prediction 2:** use laser capture microdissection to obtain matched GG1 and GG5 foci;

**Prediction 3:** perform targeted DNA sequencing for PTEN, TP53, BRCA2, and ERG status (and broader panels as needed);

# 4. Minimum Viable Validation Plan

## A. Retrospective Cohort Analysis

**Data source:** TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

## B. Tissue-Level Validation

**Methods:** Multiplex IF / IHC / spatial transcriptomics

**Key readouts:** - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

**Goal:** Confirm that the proposed mechanism operates at the spatial / tissue level.

## C. Functional Validation

**Models:** Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

**Perturbations:** Genetic (KO/rescue) or pharmacologic modulation of anchor gene

**Readouts:** T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

**Goal:** Establish causal relationship between anchor gene activity and immune efficacy.

# 5. Strengths, Caveats & Alternative Directions

## Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

## Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

## Alternative Directions